DTMBIO 2026

20th International Conference on
Data and Text Mining in Biomedical Informatics (DTMBIO)

October 6(Tue) – 10(Sat), 2026
Osaka University Nakanoshima Center, Osaka, Japan



DTMBIO

https://dtmbio.net/

DTMBIO 2026

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20th International Conference on

Data and Text Mining in Biomedical Informatics (DTMBIO)

The organizers of DTMBIO 2026 are pleased to announce the 20th International Conference on Data and Text Mining in Biomedical Informatics (DTMBIO 2026). The conference focuses on advances in biomedical informatics, genomics, and bioinformatics, bringing together researchers and practitioners to present innovative applications of state-of-the-art informatics methods in biomedical research.

Biomedical researchers are increasingly challenged to make effective use of the enormous volume of electronic biomedical data in order to better understand and explain complex biological systems. These data repositories encompass a wide range of formats, including bibliographic information from medical journals, gene expression data from sequencing experiments, protein identification and quantification data from proteomics studies, genomic sequences generated by massively parallel sequencing, and patient healthcare records. Automatically extracting, integrating, interpreting, and utilizing information embedded in such heterogeneous data—both structured and unstructured—remains a major challenge.

DTMBIO 2026 invites submissions that address these and related challenges from methodological, applied, and evaluative perspectives. Accepted full papers will be invited for a special issue of the Computational and Structural Biotechnology Journal (CSBJ) or sister journals in the Science Partner Journals (SPJ).

Topic of Interest

The relevant topics include the following (but not limited to):

- Artificial intelligence-driven analysis of large-scale biological and clinical datasets
- Protein and RNA structure prediction using deep learning approaches
- Gene and genome annotation with machine learning techniques
- Biomedical and clinical text mining and natural language processing
- Information retrieval and knowledge extraction from biomedical databases
- Genomic, transcriptomic, and multi-omics data integration and analysis
- Human microbiome data mining and modeling
- Medical ontologies and semantic data integration
- Named entity recognition and concept extraction in biomedical text
- Discovery of sequence and structural motifs
- Modeling of biochemical pathways and biological networks
- Image mining in medical and healthcare informatics
- AI-driven approaches in drug discovery, systems biology, and biomedical workflows Big bio- or clinical- data analytics

Program

6 October (Tue)

Time	Program
11:00 - 12:00	Financial and Accounting Review
12:00 - 13:00	**Lunch break**
13:00 - 14:00	Conference Organization and Governance
14:00 - 15:00	Committee Structure and Membership
15:00 - 16:00	DTMBIO 2026 Scientific Program and Keynote Speakers

7 October (Wed)

Time	Program
08:30 - 09:30	Registration
09:30 - 09:40	Welcome & Opening remark
09:40 - 10:25	**[Session I] Genomic and Regulatory Modeling (DTMBIO selected papers)** Chair: Jaejoon Choi
09:40 - 09:55	**In silico design of bacterial terminators with targeted strength using interpretable hybrid modeling** Seojin Kim (KAIST)
09:55 - 10:10	**Machine learning-based prioritization of disease-informative exon skipping events in Alzheimer's disease and progressive supranuclear palsy** Yeongmin Kim (KAIST)
10:10 - 10:25	**Multimodal Machine Learning with Cell-Specific Regulatory Features for Single-Cell DNA Methylation Prediction** Thanmayee Yeluri (Prairie View A&M University)
10:25 - 10:40	**Break time**
10:40 - 11:30	**[Keynote Talk I] Interpretable Deep Learning for Gene Regulation and Sequence Function** Peter Koo (Cold Spring Harbor Laboratory)
11:30 - 13:00	**Lunch break**
13:00 - 14:00	**[Session II] Advances in Genomics through Long-read Sequencing (Invited Speakers)** Chair: Dokyun Na
13:00 - 13:20	**Korean Pangenome Project and segmental duplications** Jun Kim (Chungnam National University)
13:20 - 13:40	**Advanced mosaic mutation detection with multi-platform sequencing technologies** Yoo-Jin Jiny Ha (Hanyang University)
13:40 - 14:00	**When One Reference Is Not Enough: Stress-Testing Long-Read Variant Analysis Beyond Diploidy** Yoshinori Fukasawa (Utsunomiya University)
14:00 - 14:30	**Break time (coffee)**
14:30 - 15:30	**[Session III] Data-Driven Systems Biology of Health and Disease (Invited Speakers)** Chair: Junho Kim
14:30 - 14:50	**Cancer Research Through the Integration of Omics and Machine Learning** Yosui Nojima (Nara Medical University)
14:50 - 15:10	**Single cell multiomics identifies transcription factors driving murine T cell development** Diego Diez (The University of Osaka)
15:10 - 15:30	**Bacterial Crosstalk in the Gut: Linking Microbial Networks to Host Health** Koji Hosomi (Osaka Metropolitan University)
15:30 - 15:50	**Break time**
15:50 - 16:40	**[Keynote Talk II] Genetic Mutations in Human Disease: From Mendelian Inheritance to Somatic Mosaicism** Sangwoo Kim (Yonsei University)
16:40 - 16:50	Closing address

Program

8 October (Thu)

Time	Program
08:30 - 09:00	Registration
09:00 - 10:20	**[Poster Session]**
10:20 - 10:30	Break time
10:30 - 11:30	**[Session I] Toxicity and Therapeutic Discovery (DTMBIO selected papers)** Chair: Jun Kim
10:30 - 10:45	**Mito-CAT: A Tri-modal Cross-Attention Network for Mitochondrial Toxicity Prediction** Jingyu Lee (Chung-Ang University)
10:45 - 11:00	**HoME-Ames: Hierarchy of Multi-Gate Mixture-of-Experts for Strain-Panel Ames Mutagenicity Prediction** Myeonghyeon Jeong (Chonnam National University)
11:00 - 11:15	**Reference-Conditioned Combinatorial Target Prioritization from Herb-Level Network Proximity Gains** Hanjin Kim (KAIST)
11:15 - 11:30	**Identifying Potential Phenotypic Effects of Natural Products and Their Origins for Alternative Source Discovery Using NPKG-GNN** Kowovi Comivi Alowonou (MATILO AI Inc.)
11:30 - 13:00	**Lunch break**
13:00 - 14:00	**[Session II] Clinical and Biomedical AI (DTMBIO selected papers)** Chair: Sunyong Yoo
13:00 - 13:15	**Subgroup Heterogeneity after Local Adaptation of an Echocardiography Foundation Model for HFrEF** Sungwon Hwang (Yonsei University)
13:15 - 13:30	**Peripheral immune marker and network associations with depressive symptoms across statistical approaches** Dawoon Kim (Chonnam National University)
13:30 - 13:45	**Lightweight CenterNet-OBB with Skeleton Auxiliary and Hardness Weighting for Chromosome Detection** Le Duc Minh Quan (Hanoi University of Science and Technology)
13:45 - 14:00	**Chronic disease classification using genetic and non-genetic health-related factors with machine learning in the Korea 4K cohort** Minkyung Choi (Seoul National University)
14:00 - 14:30	Break time (coffee)
14:30 - 15:30	**[Session III] AI-Driven Computational Biology (Invited Speakers)** Chair: Dukyong Yoon
14:30 - 14:50	**Protein analysis from multimers to metagenomes** Milot Mirdita (Sungkyunkwan University)
14:50 - 15:10	**Drug candidate discovery using machine learning** Giltae Song (Pusan National University)
15:10 - 15:30	**Advancing the Scale of Synthetic Biology using machine-learning and big data analytics** Donghui Choe (Sungkyunkwan University)
15:30 - 15:40	Break time
15:40 - 16:30	**[Keynote Talk III] Transforming Clinical and Omics Data into Actionable Knowledge for Drug Discovery and Preventive Healthcare** Yayoi Natsume-Kitatani (the Artificial Intelligence Center for Health and Biomedical Research)
16:30 - 17:00	Closing Ceremony
Banquet (18:00 - 21:00) - Folk Kitchen @ Hilton Osaka	

Program

9 October (Fri)

Time	Program
10:30 - 11:30	**RLRC workshop**
10:30 - 10:50	RLRC workshop Introduction Sunyong Yoo (Chonnam National University)
10:50 - 11:10	Digital Twin of the Cell: Multi-Omics Integration for Predictive Biology Sunyong Yoo (Chonnam National University)
11:10 - 11:30	From Multi-Omics Landscapes to Precision Medicine: Unveiling Hidden Regulatory Mechanisms of Disease Sunjae Lee (KAIST)
11:30 - 13:00	**Lunch break**
13:00 - 14:20	**K-HOPE workshop**
13:00 - 13:20	K-HOPE workshop Introduction Sunyong Yoo (Chonnam National University)
13:20 - 13:50	Single-Cell Multi-Modal Omics: Dissecting Cell-Specific Regulatory Networks in Heterogeneous Tissues Junho Kim (Sungkyunkwan University)
13:50 - 14:20	Systems Toxicology: Multi-Omics Approaches to Predict Drug-Induced Cellular Stress Jun Kim (Chungnam National University)

10 October (Sat)

Time	Program
11:00 - 12:00	Korea-Japan Collaboration in Bioinformatics
12:00 - 13:00	**Lunch break**
13:00 - 15:00	Korea–Japan Bioinformatics Expert Exchange
15:00 - 16:00	Joint Planning for Future DTMBIO Conferences

Keynote Lectures



[Keynote Lecture 1]

Interpretable Deep Learning for Gene Regulation and Sequence Function

Prof. Peter Koo

Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, USA

Peter Koo is an Associate Professor at the Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, where he develops machine learning methods to study gene regulation and biological sequence function. His research builds and interprets deep learning models that link DNA and protein sequence to molecular and cellular phenotypes, with an emphasis on interpretability, generalization, and mechanistic insight. By moving beyond black-box prediction, his work aims to produce computational frameworks that yield testable hypotheses and mechanistic understanding in regulatory genomics and cancer biology.

Keynote Lectures

[Keynote Lecture 2]



Genetic Mutations in Human Disease: From Mendelian Inheritance to Somatic Mosaicism

Prof. Sangwoo Kim

Department of Biomedical Systems Informatics, Yonsei University College of Medicine, Seoul, Republic of Korea
Graduate School of Medical Science, Brain Korea 21 Project, Yonsei University College of Medicine, Seoul, Republic of Korea

Sangwoo Kim, Ph.D. is a Professor in the Department of Biomedical Systems Informatics at Yonsei University College of Medicine, where he directs the Translational Genome Informatics Laboratory (TGIL). His research integrates computational biology, genomics, and medicine to uncover the genetic mechanisms underlying human disease through the development of innovative analytical methods and large-scale genomic studies.

Dr. Kim is internationally recognized for his contributions to somatic mosaicism and rare variant detection. He has developed widely adopted computational methods for next-generation sequencing analysis, including Virmid, one of the pioneering algorithms for somatic variant detection, and RePlow, a highly accurate framework for detecting ultra-low-frequency somatic mutations. His group has also established benchmark standards for mosaic variant calling, providing practical guidelines that have become important references for the genomics community.

His research has revealed fundamental insights into human genetic diseases and cancer, including the first identification of the genetic basis of spina bifida through de novo mutations published in Nature. More recently, his laboratory has advanced the understanding of developmental somatic mosaicism as a key mechanism underlying human disease susceptibility. His work has appeared in leading journals including Nature, Nature Methods, Nature Communications, Genome Biology, Nucleic Acids Research, Journal of Clinical Oncology, and Journal of Hepatology.

Keynote Lectures



[Keynote Lecture 3]

Transforming Clinical and Omics Data into Actionable Knowledge for Drug Discovery and Preventive Healthcare

Prof. Yayoi Natsume-Kitatani

Artificial Intelligence Center for Health and Biomedical Research, National Institutes of Biomedical Innovation, Health and Nutrition, Settsu, Osaka, Japan
Institute of Advanced Medical Sciences, Tokushima University, Tokushima, Tokushima, Japan
Institute for Protein Research, The University of Osaka, Suita, Osaka, Japan
Graduate School of Pharmaceutical Sciences, The University of Osaka, Suita, Osaka, Japan

Dr. Yayoi Natsume-Kitatani is the Director of the AI Center for Health and Biomedical Research (ArCHER) and Project Leader of the Laboratory of Bioinformatics at the National Institutes of Biomedical Innovation, Health and Nutrition (NIBN) in Japan. She earned her Ph.D. from the University of Tokyo in 2008. Following her doctorate, she was a postdoctoral fellow at the Bioinformatics Center, Institute for Chemical Research, Kyoto University, and in 2010 she became a PRESTO researcher at the Japan Science and Technology Agency. Since 2015 she has been affiliated with NIBN (formerly NIBIOHN), leading projects that apply machine learning and bioinformatics to accelerate drug discovery. She also holds concurrent appointments as a Visiting Professor at the Graduate School of Pharmaceutical Sciences and at the Institute for Protein Research, The University of Osaka. Her work centers on applying machine learning to clinical and omics data to extract actionable knowledge. She is a member of The Engineering Academy of Japan and a Fellow of the International Academy of Artificial Intelligence Sciences.

Invited Speakers

7 October (Wed) Session II

Advances in Genomics through Long-read Sequencing



Korean Pangenome Project and segmental duplications

Prof. Jun Kim
Assistant Professor
Department of Convergent Bioscience and Informatics, Chungnam National University

CAREER & EDUCATION
- Researcher in Korea Research Institute of Bioscience and Biotechnology (KRIBB)
- Postdoctoral researcher in Research Institute of Basic Sciences, Seoul National University
- Ph.D., in Biological Sciences, Seoul National University
- B.S., in Life Sciences, Pohang University of Science and Technology



Advanced mosaic mutation detection with multi-platform sequencing technologies

Prof. Yoo-Jin Jiny Ha
Assistant Professor
Department of Life Science, Hanyang University

CAREER & EDUCATION
- Research fellow in Dept. of Biomedical Informatics, Harvard Medical School
- Ph.D., in Dept. of Biomedical Systems Informatics, Yonsei University College of Medicine
- B.S., in Genetic Engineering, KyungHee University



When One Reference Is Not Enough: Stress-Testing Long-Read Variant Analysis Beyond Diploidy

Prof. Yoshinori Fukasawa
Associate Professor, Ph.D.
Utsunomiya University, Institute for Promotion of Research Center for Bioscience Research and Education
Utsunomiya University, Graduate School of Regional Development and Creativity Division of Engineering and Agriculture (Master's Course) Graduate Program in Molecular Agriculture

CAREER & EDUCATION
- Department of Computational Biology, Graduate School of Frontier Sciences, The University of Tokyo
- Department of Applied Bio-Sciences, Faculty of Agriculture, Tohoku University
- Plant biology, College of Biological Sciences, University of California, Davis

Invited Speakers

7 October (Wed) Session III

Data-Driven Systems Biology of Health and Disease



Cancer Research Through the Integration of Omics and Machine Learning

Prof. Yosui Nojima

Professor
Department of Computational Medicine, Nara Medical University

CAREER & EDUCATION
- Professor, Department of Computational Medicine, Nara Medical University, 2026-Present
- Associate Professor, Center for Mathematical Modeling and Data Science, The University of Osaka, 2023-2026
- Lecturer, Center for Mathematical Modeling and Data Science, The University of Osaka, 2020-2023
- PostDoc, Laboratory of Bioinformatics, National Institute of Biomedical Innovation, Health and Nutrition, 2018-2020
- PostDoc, Laboratory for Growth Control Signaling, RIKEN, 2017-2018
- Ph.D., Tokyo University of Agriculture and Technology, 2017



Single cell multiomics identifies transcription factors driving murine T cell development

Prof. Diego Diez

Associate Professor
Institute for Open and Transdisciplinary Research Initiatives, The University of Osaka

CAREER & EDUCATION
- The University of Osaka, WPI Immunology Frontier Research Center
- Kyoto University, Institute for Chemical Research Bioinformatics Center, Kanehisa Laboratory
- Ph.D., Biochemistry, Autonomous University, Madrid
- B.A., Biochemistry, Complutense University, Madrid



Bacterial Crosstalk in the Gut: Linking Microbial Networks to Host Health

Prof. Koji Hosomi

Associate Professor
School of Veterinary Science Department of Veterinary Science, Osaka Metropolitan University

CAREER & EDUCATION
- Associate Professor, Graduate School of Veterinary Science, Osaka Metropolitan University, 2024/09 – present
- Research Stay, Ludwig Maximilian University of Munich (LMU Munich), Germany, 2023/10 - 2024/07
- Researcher, National Institutes of Biomedical Innovation, Health and Nutrition (Senior Researcher from December 2021), 2019/01
- JSPS Postdoctoral Research Fellow (PD), 2018/04
- Specially Appointed Researcher, National Institutes of Biomedical Innovation, Health and Nutrition (Project Researcher from April 2017), 2015/04
- Ph.D. in Veterinary Medicine, Osaka Prefecture University, 2015/03

Invited Speakers

8 October (Thu) Session III

AI-Driven Computational Biology



Protein analysis from multimers to metagenomes

Prof. Milot Mirdita

Associate Professor
Sungkyunkwan University School of Medicine

CAREER & EDUCATION
- Postdoctoral researcher, Seoul National University, Seoul, Korea
- Ph.D., University of Göttingen & Max Planck Institute for Multidisciplinary Sciences
- M.S., Ludwig Maximilian University of Munich
- B.S., Ludwig Maximilian University of Munich & Technical University of Munich



Drug candidate discovery using machine learning

Prof. Giltae Song

Professor
School of Computer Science and Engineering, Pusan National University

CAREER & EDUCATION
- Ph.D., Computer Science and Engineering, Pennsylvania State University
- M.S., Computer Science and Engineering, Seoul National University, Korea
- B.S., Computer Science, Seoul National University, Korea



Advancing the Scale of Synthetic Biology using machine-learning and big data analytics

Prof. Donghui Choe

Assistant Professor
Biological Sciences, Sungkyunkwan University

CAREER & EDUCATION
- Assistant Professor, Dept. BiologicalSciences, Sungkyunkwan University, Korea
- Postdoctoral Fellow, Dept.Bioengineering, UCSD, USA
- Postdoctoral Fellow, Dept. BiologicalSciences, KAIST, Korea
- M.S. & Ph.D., Dept. BiologicalSciences, KAIST, Korea
- B.S., Dept. Biological Sciences,KAIST, Korea

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- **Feifan Liu** University of Wisconsin-Milwaukee, USA
- **Gwan-Su Yi** Korea Advanced Institute of Science and Technology, Korea
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- **Sangwoo Kim** Yonsei University College of Medicine, Korea
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Organizing Committee

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MEMO

20th International Conference on
Data and Text Mining in Biomedical Informatics (DTMBIO)

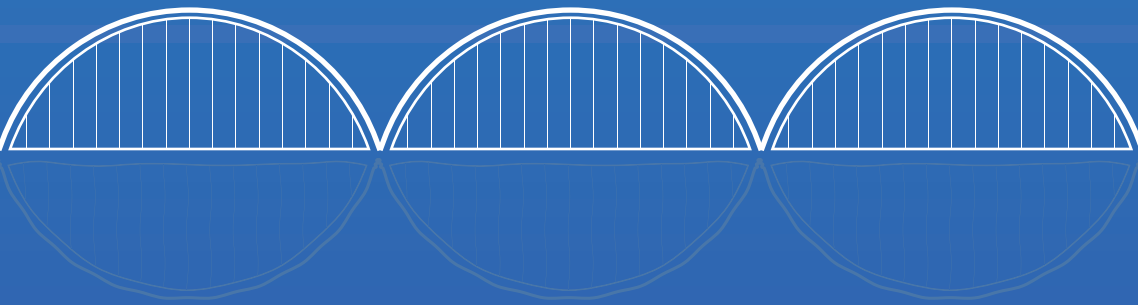
DTMBIO 2026

Selected Papers

Oct 7th (Wed) Session I
Genomic and Regulatory Modeling

Oct 8th (Thu) Session I
Toxicity and Therapeutic Discovery

Oct 8th (Thu) Session II
Clinical and Biomedical AI

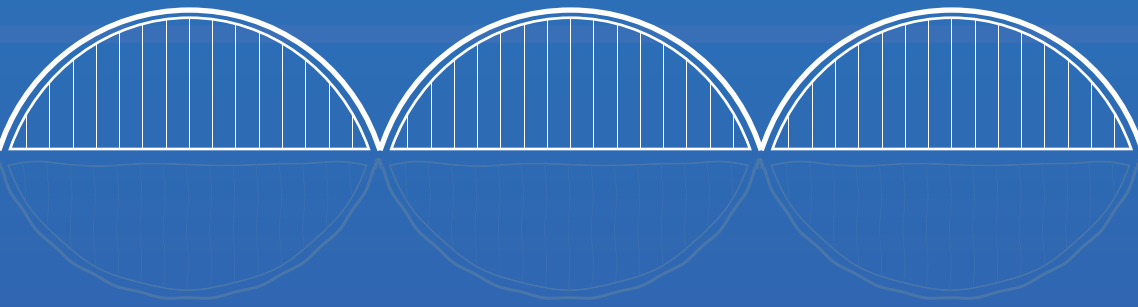


7 October (Wed) (7F A+B)

[Session I]

Genomic and Regulatory Modeling

In silico design of bacterial terminators with targeted strength using interpretable hybrid modeling	Seojin Kim (KAIST)
Machine learning-based prioritization of disease-informative exon skipping events in Alzheimer's disease and progressive supranuclear palsy	Yeongmin Kim (KAIST)
Multimodal Machine Learning with Cell-Specific Regulatory Features for Single-Cell DNA Methylation Prediction	Thanmayee Yeluri (Prairie View A&M University)



Genomic and Regulatory Modeling

7 October (Wed) Session I

In silico design of bacterial terminators with targeted strength using interpretable hybrid modeling

Seojin Kim[1] and Doheon Lee[1*]

[1]Department of Bio and Brain Engineering, Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea.
[*]Corresponding author: dhlee@kaist.ac.kr

Transcription terminators set how much of a transcript reads through into downstream genes, yet their strength still cannot be specified in advance. Computational design has so far traded one property against another: generative models explore sequence space freely but cannot account for what they produce, whereas interpretable models achieve transparency by narrowing the representation, and with it the design space. We present an in silico architecture in which interpretation instead constrains generation. A predictor trained on 582 characterized Escherichia coli terminators pairs nucleotide language model embeddings with biophysically motivated descriptors. Permutation importance and exhaustive mutagenesis of that predictor, held fixed throughout, show that strength is governed overwhelmingly by the composition and extent of the poly-T tract rather than by hairpin geometry. Its determinants fall into two short windows flanking the hairpin 3' end that respond in opposite directions. These findings are exported as explicit design rules that set both the objective and the mutation operators of an evolutionary search seeded by a genomic language model. Generation is thus driven by the interpretation rather than screened by the predictor. The framework produced designs matching each of eleven requested termination efficiencies, and an independently trained predictor reproduced the ordering of those targets, though not their absolute scale for weak terminators. Interpretation and generation can therefore be coupled rather than traded against one another, yielding terminator designs whose sequence determinants are stated explicitly and can be tested experimentally.

Genomic and Regulatory Modeling

7 October (Wed) Session I

Machine learning-based prioritization of disease-informative exon skipping events in Alzheimer's disease and progressive supranuclear palsy

Yeongmin Kim[1,†], Seonggyun Han[2,†], and Younghee Lee[3*]

[1]Department of Mathematical Sciences, KAIST, Daejeon, Republic of Korea.
[2]Department of Biomedical Software Engineering, The Catholic University of Korea, Bucheon, Republic of Korea.
[3]College of Veterinary Medicine and Research Institute for Veterinary Science, Seoul National University, Seoul, Republic of Korea.
[†]These authors contributed equally to this work.
[*]Corresponding author: amazon@snu.ac.kr

Alzheimer's disease (AD) and progressive supranuclear palsy (PSP) are distinct neurodegenerative tauopathies that share overlapping neuropathological features and transcriptomic alterations. Identifying molecular alterations that distinguish these diseases is essential for advancing our understanding of their disease-specific mechanisms and may ultimately facilitate future diagnostic studies. This study aims to uncover and prioritize biologically meaningful exon skipping (ES) signatures that differentiate between AD and PSP with a machine learning (ML) framework.
Percent-spliced-in values from a discovery cohort of 96 AD and 96 PSP cases yielded 30,799 candidate ES events; an independent cohort of 82 AD and 84 PSP cases was used for external validation. Univariate, multivariate, and embedded feature-selection methods were compared with eight classifiers across ten train-test splits.
A radial-basis-function support vector machine with F-test selection at false discovery rate < 0.05 performed best, achieving an internal area under the receiver operating characteristic curve (AUROC) of 0.901 ± 0.010 and an external-validation AUROC of 0.797 ± 0.012 (mean ± 95% confidence-interval half-width). Feature-importance analysis across kernel, tree-based, and linear models in both cohorts prioritized 234 ES events in 201 genes, enriched in neuronal, cytoskeletal, and trafficking pathways. Fifteen events ranked within the top 1% across model-cohort combinations on average; a logistic regression score based on them achieved AUROC 0.843, with repeated cross-validation yielding 0.805 ± 0.003. These findings demonstrate the utility of an ML-based framework for prioritizing disease-informative ES events and provide a focused set of candidate splicing alterations for investigating disease-specific molecular mechanisms between AD and PSP.

Genomic and Regulatory Modeling

7 October (Wed) Session I

Multimodal Machine Learning with Cell-Specific Regulatory Features for Single-Cell DNA Methylation Prediction

Thanmayee Yeluri[1] and Seungchan Kim[1*]

[1]Electrical and Computer Engineering Department, Prairie View A&M University, Prairie View, USA.
[*]Corresponding author: sekim@PVAMU.EDU

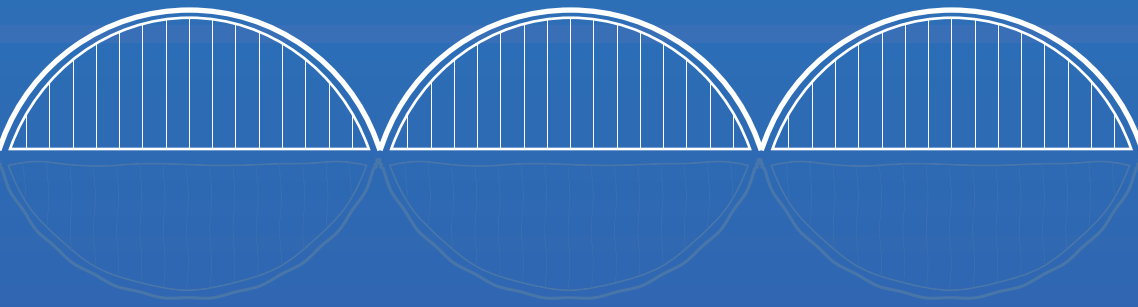
Single-cell DNA methylation measurements are often sparse, making it difficult to characterize methylation patterns across cells and genomic regions. Multi-omic measurements from the same cells provide complementary information that may help predict methylation at unobserved cell-gene pairs. Here, we developed a multimodal machine-learning framework for cell-gene DNA methylation prediction using genomic sequence, RNA expression, chromatin accessibility, transcription factor (TF) features, CpG context, and regulatory annotations. We compared XGBoost, CatBoost, a multi-layer perceptron (MLP), and a knowledge-guided neural network using a within-cell gene-split evaluation. Using the gene-associated feature representation, the MLP and Knowledge-Guided models performed similarly, with Pearson correlations of 0.6066 and 0.6072, respectively. We then incorporated cell-specific RNA, accessibility, and TF-related features to construct the full multimodal feature representation. With this representation, the MLP reached a Pearson correlation of 0.6222, whereas the Knowledge-Guided model achieved 0.6523, with RMSE = 1.2655 and R2 = 0.4249. The Knowledge-Guided model also improved cell-level correlation for 79.4% of evaluable cells and reduced prediction bias across the methylation range. Ablation experiments showed that cell-specific TF expression contributed most strongly to the improvement associated with the cell-specific regulatory information, whereas ablating TF motif, TF-expression-motif interaction, or learned gating components did not reduce performance. These results show that cell-specific molecular information can improve within-cell methylation prediction and identify TF expression as the most informative cell-specific regulatory component in the current framework.

8 October (Thu) (7F A+B)

[Session I]

Toxicity and Therapeutic Discovery

Mito-CAT: A Tri-modal Cross-Attention Network for Mitochondrial Toxicity Prediction	Jingyu Lee (Chung-Ang University)
HoME-Ames: Hierarchy of Multi-Gate Mixture-of-Experts for Strain-Panel Ames Mutagenicity Prediction	Myeonghyeon Jeong (Chonnam National University)
Reference-Conditioned Combinatorial Target Prioritization from Herb-Level Network Proximity Gains	Hanjin Kim (KAIST)
Identifying Potential Phenotypic Effects of Natural Products and Their Origins for Alternative Source Discovery Using NPKG-GNN	Kowovi Comivi Alowonou (MATILO AI Inc.)



Toxicity and Therapeutic Discovery

8 October (Thu) Session I

Mito-CAT: A Tri-modal Cross-Attention Network for Mitochondrial Toxicity Prediction

Jingyu Lee[1], Myeong-Sang Yu[1], Yunhyeok Lee[1], Kha Mong Tran[1], and Dokyun Na[1*]

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Mitochondrial toxicity remains a primary cause of organ injury and compound attrition in both drug discovery and environmental chemical safety assessments. Since experimental evaluation of this toxicity is slow and resource-intensive, several computational models have been developed to overcome this inefficiency. However, prior models relied on single data sources and single molecular representations, and a demand for high-accuracy models therefore remains. Here we introduce Mito-CAT, a tri-modal cross-attention network that predicts the mitochondrial toxicity of chemical compounds. The curated training dataset contains 18,076 compounds and is more chemically diverse than previous datasets. Each compound is represented as molecular descriptors, a molecular graph, and a chemical foundation model embedding, and Mito-CAT integrates these views through cross-attention. Under nested cross-validation, Mito-CAT achieved the highest AUC among all combinations of the three representations and three fusion strategies. Mito-CAT demonstrated a higher Matthews correlation coefficient (0.48–0.59) compared with six publicly available models when tested on three independent datasets with increasing structural dissimilarity. Furthermore, Mito-CAT maintained a high specificity (≥0.97) as well as a high positive predictive value (≥0.94), indicating that compounds flagged as toxic by the model are highly likely to be true toxicants. The curated datasets, the trained model, and a freely accessible web tool are publicly available, and these assets provide a transparent resource for the early identification of mitochondrial liabilities in environmental hazard assessments and drug development.

Toxicity and Therapeutic Discovery

8 October (Thu) Session I

HoME-Ames: Hierarchy of Multi-Gate Mixture-of-Experts for Strain-Panel Ames Mutagenicity Prediction

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The Ames test, or the bacterial reverse mutation test, is the standard assay for determining whether a chemical induces gene mutations and is required in drug development and regulatory assessment. However, since the assay is costly and labor-intensive, computational models have been explored to reduce this burden. Single-task learning models have been proposed to predict the overall judgment of mutagenicity, but they do not indicate which strain or metabolic condition drove that judgment. In contrast, multi-task learning models for strain-specific prediction can determine the overall Ames test result by combining all individual predictions. Nevertheless, jointly training multiple conditions still improves performance for some tasks while degrading it for others, and direct prediction of overall mutagenicity remains challenging. To overcome the limitations of previously proposed multi-task learning approaches, we designed a model based on a hierarchy of multi-gate mixture-of-experts (HoME-Ames), in which learnable gates group tasks by metabolic activation and strain. The model was evaluated on compounds with molecular scaffolds absent from the training data. For individual tasks, our model achieved performance comparable or superior to that of existing multi-task learning models and substantially outperformed single-task models on the condition with the fewest measurements. Overall, HoME-Ames reduces the performance degradation common in multi-task learning and provides the overall judgment together with the underlying condition-level predictions within a single model.

Toxicity and Therapeutic Discovery

8 October (Thu) Session I

Reference-Conditioned Combinatorial Target Prioritization from Herb-Level Network Proximity Gains

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Effective combination therapy depends on identifying targets that should be engaged together, but the candidate space is vast. Herbs provide naturally co-occurring sets of constituents whose targets may guide combination design. Decomposing an herb-level network signal is difficult, however: n constituents generate 2^n - 1 subsets, and overlapping targets preclude unique attribution. We therefore fixed one reference constituent within each herb-disease context and asked whether a named partner and its added targets could recover disease-module exposure beyond the reference and capture part of the herb-level proximity gain. Across 3,726 herbs, 415 constituents with experimentally supported targets, and 44 disease modules in the human interactome, constituent-mediated clinical-trial evidence was enriched among proximal herb-disease pairs and among herb-disease contexts with large herb-reference gaps, beyond target-module size (odds ratios 2.15 and 2.44; size-adjusted 1.93 and 1.58). Robust gap screening yielded 97 partner-specific added-target sets, represented by 530 herb-supported proposals across 15 cancer-related diseases. The resulting pairs showed complementary exposure when the reference was already exposed (23% of proposals) and rescued exposure when it was not (68%); in rescue proposals the median z-score shifted from +0.71 for the reference to -1.42 for the pair, while size-matched random target additions reached only +0.33. Real pairs beat all 10 degree-matched random additions in 90.4% of proposals. A potency-restricted rerun reproduced these patterns, supporting application to constituents without measured potency. This reference-conditioned framework narrows an intractable subset search to experimentally testable pair-target hypotheses without claiming unique attribution or synergy.

Toxicity and Therapeutic Discovery

8 October (Thu) Session I

Identifying Potential Phenotypic Effects of Natural Products and Their Origins for Alternative Source Discovery Using NPKG-GNN

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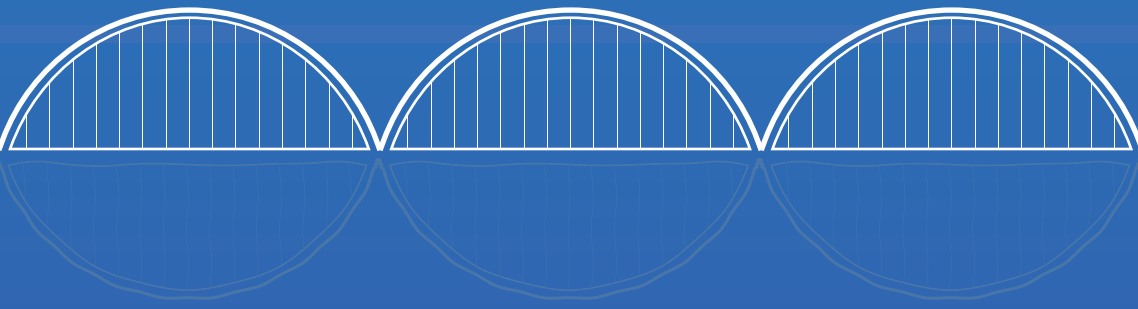
Natural Product Origins (NPOs), including herbs and other botanical raw materials, are widely used as functional ingredients because their constituent Natural Products (NPs) exhibit diverse biological activities. However, safety concerns associated with some commonly used NPOs and the need to broaden the range of available functional raw materials have increased interest in identifying alternative sources with comparable biological effects. In this study, we developed a knowledge graph-based framework to predict potential phenotypic effects of NP and their origins and to explore candidate alternative NPOs. A disease-centered heterogeneous knowledge graph comprising 155,091 entities, approximately 4.06 million relationships, and 87 semantic relation types was constructed by extending and harmonizing data from NPASS, CMAUP, COCONUT, the Drug Repurposing Knowledge Graph, and the Comparative Toxicogenomics Database. The graph connected NPs, NPOs, genes and proteins, and phenotypes through constituent, molecular interaction, and phenotype association relationships. We developed a Mechanism-Aware Disease-Profile GNN consisting of a three-layer CompGCN encoder and a ComplEx decoder to learn relation-aware entity representations and predict potential phenotype associations. For NPO-level prediction, phenotype information derived from constituent NPs and their associated genes was integrated through an adaptive gating module. A GRU-based path scorer further evaluated biological paths connecting NPOs to phenotypes. The model generated ranked disease phenotype predictions for NPs and NPOs, enabling candidate alternatives to be prioritized based on similarity to the predicted disease profiles of reference NPOs. This framework enables the prediction of potential phenotypic effects for NPs and NPOs with limited existing phenotypic information and supports the prioritization of candidate alternative NPOs based on their predicted disease profiles and underlying molecular relationships.

8 October (Thu) (7F A+B)

[Session II]

Clinical and Biomedical AI

Subgroup Heterogeneity after Local Adaptation of an Echocardiography Foundation Model for HFrEF	Sungwon Hwang (Yonsei University)
Peripheral immune marker and network associations with depressive symptoms across statistical approaches	Dawoon Kim (Chonnam National University)
Lightweight CenterNet-OBB with Skeleton Auxiliary and Hardness Weighting for Chromosome Detection	Le Duc Minh Quan (Hanoi University of Science and Technology)
Chronic disease classification using genetic and non-genetic health-related factors with machine learning in the Korea 4K cohort	Minkyung Choi (Seoul National University)



Clinical and Biomedical AI

8 October (Thu) Session II

Subgroup Heterogeneity after Local Adaptation of an Echocardiography Foundation Model for HFrEF

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Transfer learning is the default recipe for adapting foundation models to new clinical sites. While this local adaptation is often treated as a technical safeguard for clinical deployment, it can also become a governance blind spot. We argue that an aggregate area under the receiver operating characteristic curve (AUROC), even when fully recovered, can hide clinically meaningful subgroup-level performance heterogeneity in deployment-time performance. Using EchoPrime on 2,984 MIMIC-IV-ECHO patients, we validated detection of left ventricular ejection fraction (LVEF) ≤ 40% - the threshold gating care for heart failure with reduced ejection fraction (HFrEF). Pooled transfer learning restored aggregate AUROC to 0.905 (95% Confidence Interval 0.853–0.950). A subgroup-stratified audit revealed performance heterogeneity across all examined axes. AUROC ranged from 0.858 to 0.924 across eight subgroups with the lowest value observed in patients with low Charlson Comorbidity Index group (CCI ≤ 2). Subgroup miss rates also diverged: CCI ≥ 3 missed 50.0% of positives versus 28.6% in CCI ≤ 2 (gap of 21.4 percentage points), with parallel disparities along sex (male: 42.9% versus female: 46.7%), age (≥ 65: 45.2% versus < 65: 37.5%), and body mass index (≥ 30: 52.9% versus < 30: 39.4%). Our findings contribute to ongoing discussions about responsible AI in clinical deployment, particularly in demonstrating that post-adaptation subgroup audits are essential for ensuring consistent detection performance across patient strata, since subgroup miss-rate disparities translate directly into unequal downstream care.

Clinical and Biomedical AI

8 October (Thu) Session II

Peripheral immune marker and network associations with depressive symptoms across statistical approaches

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Depressive symptoms have been linked to changes in the immune system, but reported biomarkers vary across studies and may depend on how data are analysed. We asked whether peripheral immune differences associated with depressive symptoms remain stable across statistical approaches. Participants were grouped by Patient Health Questionnaire-9 (PHQ-9) scores, and blood samples were analysed for 121 immune-cell markers and 10 plasma cytokines. The primary analysis adjusted for age, sex, body mass index, smoking, and medical comorbidity. Robustness was then tested with alternative adjustment, matching, weighting, sample restriction, and repeated subsampling strategies. Sixteen markers including 15 cell markers and interleukin-1β showed nominal group differences. Among those, 15 associations retained the same direction across all 28 estimable analytical conditions. Natural killer cell and CD4 T-cell markers were among the most consistent findings. Network analysis also identified one relationship between immune-cell markers that remained significant after multiple-testing correction, with marginal evidence of a broader change in correlation structure. These results suggest that depressive symptoms are associated with selected immune features that are directionally consistent across analytical choices, but further studies are needed to confirm their biological and clinical relevance.

Clinical and Biomedical AI

8 October (Thu) Session II

Lightweight CenterNet-OBB with Skeleton Guidance and Hard-Sample Weighting for Chromosome Detection

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Detecting individual chromosomes and accurately estimating their orientations is an initial step in automated karyotype analysis. This task remains challenging because chromosomes are elongated, slender, arbitrarily oriented, and often touching or overlapping. We propose a lightweight anchor-free CenterNet-OBB detector. The model represents each chromosome by its center point, uses the covariance component of the Gaussian Wasserstein Distance (GWD) to jointly learn size and orientation, and enhances representation through three complementary mechanisms: hard-sample weighting based on slenderness and overlap level, instance-specific skeleton supervision generated independently for each instance, and selectively placed DCNv2 in the deep feature stream and multi-scale feature fusion. Patient-level 5-fold experiments use 606 images from 49 patients in AutoKary2022 after removing a duplicated patient from the distributed dataset package. The proposed model achieves an AP50 of 96.375 ± 0.626%, Precision of 97.225 ± 0.608%, Recall of 95.173 ± 1.380%, F1 of 96.183 ± 0.801%, and 1.164 ± 0.255 false positive predictions per image. The model simultaneously achieves a matched-pair IoU of 85.883 ± 0.512%, a center error of 2.671 ± 0.174 pixels, and an angular error of 3.306 ± 0.147°. These results indicate that the proposed architecture produces accurate OBB outputs with few false detections, making it suitable as an input for subsequent chromosome separation and classification steps.

Clinical and Biomedical AI

8 October (Thu) Session II

Chronic disease classification using genetic and non-genetic health-related factors with machine learning in the Korea 4K cohort

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Chronic non-communicable diseases represent a growing global health burden, with multiple factors, including genetic predisposition and diverse non-genetic characteristics, contributing to disease risk. Although genetic and non-genetic factors have each been investigated for disease prediction, their relative and integrative contributions to predicting chronic diseases remain unclear. Here, we evaluated Light Gradient Boosting Machine (LightGBM) classifiers for nine chronic diseases, including obesity, diabetes, and hypertension, using matched whole-genome sequencing (WGS) and self-reported questionnaire data on 16 non-genetic health characteristics from the Korea 4K cohort (n=2,317). We compared three approaches: 1) non-genetic characteristics alone, 2) genetic features alone, and 3) an integrated model combining both in order to investigate a potential power of non-genetic factor as a chronic disease risk. The non-genetic-only models achieved AUROCs of 0.64–0.81 across the nine diseases, with the highest performance observed for diabetes (0.81) and obesity (0.78), whereas the genetic-only models performed close to chance level (AUROC, 0.47–0.54). The integrative models did not improve predictive performance over the non-genetic-only models. Moreover, SHAP-based feature importance analysis identified comorbidity history (excluding the target disease), family history, and dietary habits as the most important non-genetic features in the non-genetic-only and integrative models. In summary, our findings suggest that non-genetic characteristics were more predictive of chronic diseases than genetic features, with no additional predictive improvements from integrating genetic information. These findings support the potential of survey-based, low-cost screening and highlight modifiable non-genetic characteristics as potential targets for chronic disease prevention.

8 October (Thu) (7F C)

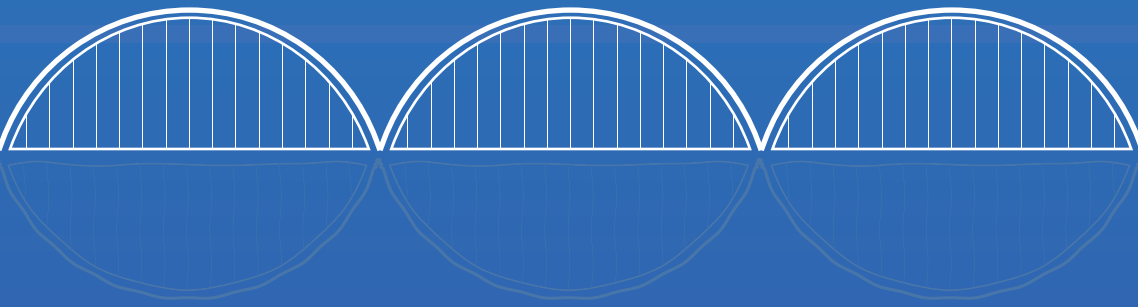
Poster Abstracts

Session A

Genomics, Epigenomics & Sequencing Data Science

Session B

AI & Modeling for Clinical and Pharmacological Research



Session A-01

Dropbucket: Robust genetic demultiplexing for low-abundance donor recovery

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Multiplexed single-cell RNA sequencing from multiple donors is an effective strategy to reduce batch effects and lower experimental costs. Existing genotype-based methods reconstruct the donor of origin for each cell but can be sensitive to sample pooling ratio and sequencing depth. Here we present dropbucket, a robust demultiplexing method that employs a weighted fuzzy mixture model to enable accurate cell assignment even under extreme pooling conditions. Using simulated pooled datasets derived from real data, we show that dropbucket outperforms existing demultiplexing methods not only under baseline pooling conditions but also extreme pooling conditions. We further demonstrate its utility in maternal–fetal interface and donor–recipient transplant datasets, highlighting the broad applicability for genetic demultiplexing scenarios.

Session A-02

ONT-only genome assembly of a Korean male individual using a semen sample

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Long-read sequencing has enabled the generation of high-quality human genome assemblies, but many previous assemblies were based on blood-derived DNA and often relied on limited data types from a single sequencing strategy. This study aimed to generate high-quality phased genome assemblies of a Korean individual using multiple independent long-read datasets produced from a single sequencing platform and to evaluate their utility for chromosome-scale assembly and variant detection. Genomic DNA was extracted from a semen sample of a Korean male. Long-read, ultra-long-read, and chromatin conformation capture sequencing data were generated using Oxford Nanopore Technologies. These datasets were integrated to construct phased genome assemblies, followed by correction of noticeable phasing errors and assessment of assembly continuity, chromosomal representation, telomeric repeat recovery, and variant detection performance. The final phased assemblies spanned approximately 2.9 Gb and represented 23 pairs of chromosomes with an NG50 of 150 Mb. Telomeric repeats were detected at 36 and 37 of the 48 chromosomal ends in the two assemblies, indicating high end-to-end completeness. In addition, we successfully identified structural variants, including small variants. These results demonstrate that combining multiple Oxford Nanopore data types can produce highly continuous and informative phased human genome assemblies. We generated high-quality phased genome assemblies of a Korean individual using Oxford Nanopore long-read sequencing data derived from semen DNA. This publicly available genome resource will support broader applications of long-read sequencing in human genomics and variant analysis.

Session A-03

MethylDC deconvolves cell types at single molecule resolution from bulk long-read sequencing data

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DNA methylation is a key epigenetic determinant of cell identity and forms stable, cell-type-specific signatures. In the mammalian brain, these signatures span both CpG (mCG) and non-CpG (mCH) contexts, with mCH accumulating in neurons during postnatal maturation and persisting throughout life. PacBio HiFi long-read sequencing provides highly accurate genome sequencing while directly capturing DNA methylation from native DNA at single-molecule resolution. However, no existing method assigns the cellular origin of individual reads carrying somatic mutations from bulk long-read sequencing data. Here, we present MethylDC, a random forest–based framework that deconvolves the cell-type composition of bulk PacBio HiFi sequencing reads by leveraging single-cell-based methylomes to assign individual reads to their cell of origin. By integrating read-level methylation patterns with somatic variants in human tissues, MethylDC enables cell-type-resolved interpretation of bulk long-read data. MethylDC is developed by using PacBio HiFi data of three human cell lines, and performance enhanced with normal brain tissues. This approach has a potential to broaden the application of long-read sequencing by enabling cell-type-specific interpretation of genetic and epigenetic information for disease research and precision medicine.

Session A-04

Somatic mosaicism in motor cortex neurons delineates a subgroup of sporadic ALS with advanced TDP-43 pathology

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Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disorder characterized by progressive motor neuron loss and cytoplasmic aggregation of phosphorylated TDP-43 (pTDP-43). Most ALS cases occur sporadically, exhibit substantial genetic and molecular heterogeneity, and lack clearly defined genetic contributors to pathogenesis. To investigate whether clonal somatic mosaicism in motor neurons contributes to sporadic ALS (sALS), we performed whole-genome sequencing of laser-captured motor cortex neurons with pTDP-43 aggregates from 26 patients with sALS, alongside pTDP-43 aggregation-negative neurons from 8 neurotypical controls. Integrative analysis of somatic mutations, transcriptomic profiles, and clinicopathological features identified a distinct genomic subgroup of sALS, termed ALS-high (13 of 26 cases), characterized by a significant increase in somatic single-nucleotide variants enriched for mutational signatures of oxidative damage. These mutations were acquired during neuronal development and were detected in both pTDP-43 aggregation-positive and -negative motor neurons. The ALS-high subgroup further corresponded to a known molecular subtype of sALS with oxidative stress-associated transcriptomic profiles and exhibited more rapid progression of pTDP-43 aggregation, suggesting that oxidative stress-induced somatic mutations arising during neuronal development predispose motor cortex neurons to pTDP-43 pathology. Together, our findings indicate that clonal somatic mosaicism in motor neurons contributes to selective vulnerability to pTDP-43 pathology and underlies molecular heterogeneity in sALS.

Session A-05

Haplotype-resolved genome assembly resolves the modular structure and gene defects of the RCCX locus in congenital adrenal hyperplasia

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The RCCX locus, resulting from segmental duplication, exhibits extensive sequence identity and modular variations because of unequal crossover events, leading to copy number variations and the formation of chimeric genes between active and pseudogenes. Precise characterization of this locus is essential for molecular diagnosis, because aberrations within this region can cause congenital adrenal hyperplasia (CAH), an autosomal recessive disorder. However, the intricate modular structures, comprising mono-, bi-, and tri-modular haplotypes, have posed challenges to accurate analysis using conventional sequencing technologies. In this study, we generated high-fidelity long-read sequencing data and constructed 26 phased genome assemblies from 13 Korean patients with CAH, achieving single-nucleotide resolution analysis of the RCCX locus. Furthermore, by integrating human draft pangenome reference data, we analyzed non-patient cohorts and identified characteristic variations unique to the Korean patient population and novel structural variants formed through DNA damage and homologous recombination. Our findings provide a key methodological framework for resolving segmentally duplicated genomic regions associated with genetic disorders. This study is expected to serve as a foundation for the precise molecular diagnosis of patients with CAH.

Session A-06

Tissue-Dependent DNA Methylation Changes in Age-Related Macular Degeneration

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Age-related macular degeneration (AMD) is a leading cause of visual impairment, yet tissue-specific epigenetic alterations in the human neuroretina and retinal pigment epithelium (RPE)/choroid remain incompletely understood. We characterized DNA methylation patterns in postmortem neuroretina and RPE/choroid tissues from two AMD donors and one healthy control donor, complemented by publicly available control datasets. Multidimensional scaling analysis revealed clear segregation between the neuroretina and RPE/choroid, demonstrating distinct tissue-specific methylation profiles. In contrast, paired left and right eyes from the same donors showed no significant methylation differences despite differences in disease stage and pathological severity. Tissue-stratified analyses at both CpG-site and regional levels identified AMD-associated methylation alterations, with substantially more differentially methylated loci in the RPE/choroid than in the neuroretina. These alterations were enriched in genes and pathways related to visual function and AMD. Notably, methylation changes were observed in PDE6B, RBP3, ABCA4, and STRA6, genes involved in phototransduction, retinoid transport, and visual cycle regulation. Together, our findings reveal distinct tissue-specific methylation signatures associated with AMD and highlight more extensive epigenetic alterations in the RPE/choroid.

Session A-07

LongMosaic: Hybrid Learning Enables Ultra-Low-Frequency Mosaic Variant Detection in Long-Read Sequencing

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Postzygotic mosaic variants are typically present at variant allele frequencies (VAFs) below 10%, making accurate detection inherently challenging. Combined with the relatively high sequencing error rate of long-read sequencing, ultra-low-frequency mosaic variants (<5%) are difficult to distinguish from sequencing artifacts. Consequently, existing long-read variant callers often exhibit limited sensitivity for single nucleotide variants (SNV) below 5% VAF, limited accuracy in detecting small INDELs, and degraded performance in difficult genomic regions. Here, we present LongMosaic, a hybrid learning framework for ultra-low-frequency mosaic variant detection from long-read sequencing data. LongMosaic combines a pileup-based convolutional neural network (CNN), which learns platform-specific sequencing error patterns, with a feature-based Random Forest classifier that integrates read-level features to distinguish true mosaic variants from artifacts. By separating pattern learning from feature-based refinement, LongMosaic simultaneously improves sensitivity for ultra-low-frequency variants while maintaining high precision. Across synthetic mixture datasets, independent benchmark datasets, and real tissue samples, LongMosaic consistently outperformed existing long-read mosaic variant callers. In particular, LongMosaic achieved substantially higher sensitivity for variants below 5% VAF while maintaining comparable precision. Also, we observed improved performance of detecting small INDELs, especially in difficult genomic regions. The framework maintained consistent performance across diverse sequencing depths and wide variant allele frequency range Together, these results demonstrate that LongMosaic enables sensitive and accurate mosaic variant detection from long-read sequencing data and provides a practical framework for studying somatic mosaicism in both research and clinical settings.

Session A-08

Long-Read Sequencing Reveals Age-Related Isoform Diversity in African turquoise killifish

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The African turquoise killifish (Nothobranchius furzeri), the shortest-lived vertebrate model for aging research, was used to investigate isoform-level transcriptome diversity across three key life stages (3, 8.5, and 14 weeks), with a focus on alternative splicing changes in genes exhibiting prominent expression shifts during aging. PacBio Iso-Seq data were generated from pooled muscle and brain samples across three age groups, and transcript isoforms were classified using the FLAIR2 and SQANTI3 pipelines. Based on the long-read data, we identified a total of 68,101 isoforms, of which 22,292 corresponded to alternatively spliced forms absent from the reference annotation. Gene Ontology enrichment analysis of genes associated with age-related expression changes revealed enrichment of the protein phosphorylation pathway, and in-depth analysis of up-regulated transcripts identified a progressive increase in intron retention in CDKL1 during aging. These findings suggest that splicing transitions in the transcriptome may serve as a key molecular feature of the aging process.

Session A-09

The Intestinal Epithelium in Crohn's Disease: Somatic Mutational Landscape and Clinical Implications

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Chronic inflammation contributes to somatic mutation accumulation in the intestinal epithelium, yet their functional impact on Crohn's disease (CD) remains unclear. This study investigated the mutational landscape and their clinical implications for disease progression and treatment response. Deep whole-exome sequencing was performed on 219 tissues across four inflammation statuses, including longitudinal follow-up. We observed that chronic inflammation significantly increased mutation burden, with CD exhibiting a distinct mutational profile from ulcerative colitis (UC), in cytokine signaling pathways. Beyond broad mutations affecting epithelial homeostasis, those in CD-specific cytokine signaling (IL-17A, TNF-α, and IFN-γ) were notably prevalent in active-inflamed tissues. Mutations identified in NF-κB-related pathways suggest that cytokine signaling mutations affect homeostasis via NF-κB activation. In particular, small insertions in RIPK2 were recurrently identified in CD remission tissues, suggesting loss-of-function in an upstream activator of NF-κB that attenuate inflammatory signaling. Evaluation of the clinical impact of these mutations in patients receiving anti-TNF-α therapy revealed that mutations in TNF-α-related pathways significantly lowered the probability of remission. Overall, our study demonstrates somatic mutations affect both epithelial homeostasis and the persistence of chronic inflammation. These findings suggest that integrating somatic mutational profiling into clinical practice may facilitate personalized treatment strategies for patients with non-responsive IBD.

Session A-10

Butyrate Producers, Enterobacteriaceae, and Bile-Acid Deconjugation Associate with Insulin sensitivity in Drug-Naïve Older Adults

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Insulin resistance in older adults has been linked to gut microbial metabolism, but most cohorts include participants on glucose-lowering therapy, which confounds this relationship. In 110 treatment-naive older Korean adults (45 men, 65 women; mean age 72.1 years; 40 normal glucose tolerance, 29 prediabetes, 41 type 2 diabetes), we profiled the gut microbiome by shotgun metagenomics (1,379 metagenomic species pangenomes [MSPs]; IGC2/Meteor2) and the serum metabolome by targeted metabolomics (Biocrates MxP Quant 500; 222 metabolites after QC). Insulin sensitivity was estimated with a Matsuda index modified for a three-point OGTT (0, 60, 120 min). Associations were assessed by age- and sex-adjusted linear models and partial Spearman correlations (significance at FDR<0.05), and by GSEA-based taxon-set enrichment. A butyrate-producing guild (151 MSPs encoding a complete acetyl-CoA pathway to butyrate, including the terminal but or buk genes) and Enterobacteriaceae showed opposing associations with insulin sensitivity (partial ρ=+0.27 and −0.33), whereas alpha diversity (Shannon, richness) was not associated with the Matsuda index. Enrichment was concentrated in insulin-axis traits (Matsuda index, fasting insulin, C-peptide) rather than in glycemic traits (fasting glucose, HbA1c). In metabolome-integrated analyses, the butyrate guild was inversely correlated with taurochenodeoxycholate (TCDCA; ρ=−0.30), which was itself inversely associated with insulin sensitivity (ρ=−0.37). Bile-salt-hydrolase–encoding taxa co-occurred with butyrate producers and were depleted in participants with higher Enterobacteriaceae abundance; adjusting for these taxa attenuated the guild–insulin-sensitivity association. In this cross-sectional, medication-free cohort of older adults, a butyrate-producer/Enterobacteriaceae axis and microbial bile-acid deconjugation tracked insulin sensitivity, identifying a taxonomically resolved microbial signature of insulin resistance.

Session A-11

A Normative Modeling Framework for Cell-free RNA

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Cell free RNA (cfRNA) liquid biopsy presents an inherently low signal to noise ratio due to substantial background signals and technical variability arising from preanalytical and experimental factors. We confirmed these limitations through a comprehensive variance analysis across multiple public datasets, which revealed that technical variation and batch effects consistently outweighed the variance associated with disease phenotypes. Particularly in the context of the high heterogeneity of complex diseases, conventional analyses comparing cases and controls based on mean values often obscure signals unique to each patient and introduce considerable instability in mean estimation within small cohorts. To address these structural limitations, we propose a normative modeling framework tailored for cell free messenger RNA(cf-mRNA) data derived from whole transcriptome sequencing. This approach utilizes multidimensional technical bias metrics derived directly from sample data as covariates to model the conditional expression distribution of normal controls. By quantifying the statistical deviation of individual samples from this reference distribution, cross validation confirmed that the deviation scores of the model were properly calibrated on unobserved normal samples, and single batch exclusion evaluation further demonstrated that the framework statistically distinguishes disease from normal samples even in batches unseen during training. Furthermore, while individual deviations were heterogeneous at the gene level, we observed a consistent convergence of these signals at the biological pathway level. This finding suggests that minimizing signal cancellation at the group level may improve the detectability of biological pathways associated with diseases in individually heterogeneous data. Ultimately, this study provides a reliable analytical alternative for integrating heterogeneous public data to discover precise biomarker candidates for individual patients.

Session B-01

A Multi-Task Learning Approach with Multiple Molecular Representations to Predict Cytochrome P450 Isoform Inhibitors

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Cytochrome P450 (CYP450) enzymes mediate phase I drug metabolism, and inhibition of major isoforms can alter drug exposure, leading to toxicity or reduced efficacy. However, existing CYP450 inhibition prediction models often focus on single isoforms, overlook inter-isoform relationships, and provide limited structural interpretability. Here, we propose a multi-task learning-based multi representation molecular framework (MTMM) for simultaneous and interpretable prediction of inhibitors across five major CYP450 isoforms. MTMM integrates graph attention, squeeze-and-excitation convolution, and transformer modules to capture complementary molecular representations, which are combined through weighted feature fusion and passed to isoform-specific prediction heads. MTMM outperformed a previous approach across all five isoforms, achieving AUROC values of 0.917–0.988 and AUPRC values of 0.790–0.989. Grad-CAM analysis enabled substructure-level interpretation, highlighting motifs associated with CYP450 inhibition, including piperazine, quinoline, and succinic acid. To evaluate practical utility, virtual screening of a ChEMBL-derived drug set prioritized three azole antifungals— isoconazole, tioconazole, and luliconazole (LCZ)—as candidate inhibitors. In vitro assays confirmed reduced CYP450 activity across isoforms, with LCZ showing the strongest time- and NADPH dependent inhibition, particularly against CYP3A4 and CYP1A2. Molecular docking and dynamics simulations further supported stable heme-proximal binding of LCZ to both isoforms. Overall, MTMM provides an interpretable and experimentally validated pipeline for early detection of compounds with complex CYP450 inhibition liabilities.

Session B-02

Clinical Research Automation through Structured Research Design and Data Analysis

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Translating a clinical question into an analysis of electronic medical record (EMR) data requires schema knowledge that clinical researchers may not have. In research EMRs one concept is stored across multiple tables. Researchers thus spend substantial time locating the relevant column. A model settling data location and aggregation on its own can produce an unintended analysis. We developed MEGIQ (Medical Evidence Generation Integrating AI-based Data Quality and Uncertainty Quantification) to map a clinical question onto the schema. We examined three capabilities. First, whether it decomposes a question into explicit design items. Second, whether it links clinical expressions to columns that exist in the schema. Third, whether it presents traceable candidates and computation paths for variables that are ambiguous or absent as a single column. MEGIQ couples a closed-network Large Language Model (LLM) with deterministic schema retrieval. A 4-bit quantized Qwen3-235B model ran on a closed-network cluster of eight Mac Minis. The LLM performs interpretation and conceptual reasoning, and the schema index verifies that a column exists. The design path decomposed each question into seven items, leaving unstated items blank. For concepts absent as a single column, the system proposed component variables and a formula. Each component was then confirmed against the index. Scenarios were applied to MIMIC-IV v3.1. We checked each output against the question. One scenario asked for the association between serum albumin at ICU admission and 90-day mortality in adults with acute kidney injury, by logistic regression adjusted for age, sex, and SOFA score. Of the seven items, the system filled in analysis type, inclusion criteria, follow-up period, and variables. The remaining items were left blank rather than inferred. BMI was absent as a single column. The system identified weight and height, confirmed both in the schema, and presented the computation path. The system does not settle unstated design items or ambiguous column mappings on its own. It returns them to the researcher with the evidence behind each candidate. This keeps the points requiring judgment visible, so the researcher decides them rather than the model. The system also runs entirely within the institution, where commercial APIs are unavailable.

Session B-03

Tumor Microenvironment Aware Transfer Learning Model for Cancer Drug Response Prediction

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Cancer is a highly complex and heterogeneous disease, and patients can exhibit different responses to the same anticancer drug. Drug response is influenced not only by cancer type and stage but also by patient-specific factors such as the tumor microenvironment (TME). Therefore, accurately predicting cancer drug response (CDR) at the patient level is crucial for advancing precision oncology. However, clinically labeled drug response data remain limited, whereas large-scale drug response datasets are relatively abundant at the preclinical level, particularly in cell lines. For this reason, many studies have attempted to leverage knowledge from preclinical data for patient level prediction using transfer learning approaches, including domain adaptation. Nevertheless, many existing methods primarily focus on learning shared representations between cell lines and patients, which may limit their ability to capture patient-specific characteristics. In this study, we propose a transfer learning-based framework for patient-specific CDR prediction that incorporates patient-specific TME information. The proposed framework consists of two branches: a preclinical knowledge transfer branch and a TME branch. The preclinical knowledge transfer branch fine-tunes a CDR prediction model pretrained on large-scale cell line data, while the TME branch independently learns representations from TME information. The outputs of the two branches are combined to predict patient-level CDR. The proposed model achieved a maximum AUROC of 0.80 and a maximum AUPR of 0.92. To evaluate the contribution of the TME branch, we further conducted an ablation study. The model with TME information generally showed better performance than the model without TME information. These results suggest that combining transferred preclinical knowledge with patient-specific TME information can improve patient-level CDR prediction. A key contribution of this study is the incorporation of patient-specific TME information, which has been relatively overlooked in previous preclinical-to-clinical knowledge transfer approaches. The proposed framework considers both shared representations across domains and patient-specific TME features, thereby contributing to more precise and personalized cancer treatment.

Session B-04

OmicsDRP: Interpretable Multi-Omics and Molecular Representation Learning for Drug Response Prediction

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Precision medicine aims to identify optimal treatments based on the molecular profiles of individual patients. However, predicting drug response remains a major challenge due to intratumoral heterogeneity. To fully capture this biological complexity, integrating comprehensive multi-omics profiles has become essential, driving the necessity for methodologies that can leverage complex biological and chemical data modalities. In this study, we propose OmicsDRP, a dual-branch deep learning model that predicts drug response by integrating cellular multi-omics profiles and drug structures. This architecture enables each branch to learn informative representations from distinct data modalities, thereby effectively capturing cross-modality interactions governing drug sensitivity. Through cross-validation on the Genomics of Drug Sensitivity in Cancer (GDSC) dataset, OmicsDRP demonstrates consistent improvements in predictive performance over conventional machine learning methods and exhibits competitive predictive accuracy compared to existing drug response prediction models. Furthermore, comprehensive interpretability analyses—including latent embedding visualizations and feature-level SHAP analysis—reveal biologically meaningful insights aligned with known drug targets and mechanisms of action, validating the model's reliability at the molecular level. In summary, OmicsDRP presents a reliable framework for precision oncology by combining robust predictive performance with biological interpretability.

Session B-05

Assessing ECG Representation-Based Drift Monitoring Across Distribution Shifts in Post-Deployment Environments

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Medical artificial intelligence (AI) models may encounter changes in patient characteristics and clinical environments after deployment, potentially affecting predictive performance. Continuous monitoring is therefore important for reliable clinical AI. For ECG-based AI models using raw waveforms, foundation-model embeddings may provide a practical way to monitor distribution shifts across clinical environments. This study aimed to evaluate whether ECG foundation-model representations can support the detection of clinically relevant distribution shifts after model development. We further examined whether representation-level changes differed across deployment scenarios and whether these changes were accompanied by deterioration in downstream predictive performance. Linked MIMIC-IV-ECG and MIMIC-IV data were used to construct development and deployment datasets under multiple simulated clinical shifts, including temporal and demographic shifts. 12-lead ECGs were transformed into 1,024-dimensional embeddings using ECGFounder. The embeddings were used to develop downstream models for predicting the presence of a cardiac condition and a laboratory measure relevant to cardiac function. Distributional differences between development and deployment representations were quantified using sliced Wasserstein distance (SWD), with a reference threshold derived from variation within the development data. Monitoring patterns varied across clinical shifts and prediction tasks. Several deployment scenarios showed noticeable representation-level changes together with reduced predictive performance, whereas others showed limited distributional changes or preserved performance. Performance deterioration was also observed in some cases without the representation distance exceeding the predefined threshold. These findings suggest that ECG foundation-model representations can provide a useful signal for identifying changes in clinical deployment environments. However, representation-level monitoring alone may not capture all forms of model deterioration. Combining representation-based distribution monitoring with performance assessment may therefore support more reliable post-deployment surveillance of ECG-based medical AI.

Session B-06

CKD2ESRD: A Hierarchical Deep Learning Approach for Interpretable ESRD Progression Prediction

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Chronic kidney disease (CKD) can progress to end-stage renal disease (ESRD), requiring dialysis or kidney transplantation. However, many existing prediction models rely on information from a single time point and provide limited insight into how longitudinal clinical records shape individual risk. We therefore developed CKD2ESRD, an interpretable model that predicts ESRD progression and identifies the clinical records contributing to each prediction. We used National Health Insurance Service claims and health-screening data from 2013 to 2023, including 171,556 patients with CKD, of whom 10,759 (6.2%) developed ESRD. Diagnosis, procedure, and drug codes were represented using embeddings from a pretrained language model and combined with health-screening variables. CKD2ESRD uses hierarchical attention across codes within each visit and across visits over time. The model predicted 3-year ESRD risk and estimated time to ESRD while accounting for censoring and death as a competing event. CKD2ESRD achieved an AUROC of 0.967 (95% CI, 0.959–0.974) and an AUPRC of 0.603, outperforming longitudinal EHR and conventional machine-learning models. Predicted risk groups showed distinct cumulative ESRD incidence after accounting for death. Patient-level heatmaps identified when important clinical events occurred and which codes contributed to each prediction, enabling the examination of patient-specific temporal patterns before ESRD progression. Removing highly ranked codes reduced predicted risk more than random deletion, supporting the faithfulness of the attention-based explanations. Overall, CKD2ESRD combined strong predictive performance with patient-level interpretation of longitudinal clinical data. By identifying both when risk-related events occurred and which records contributed to the prediction, the model may support earlier risk stratification and clinical decision-making.

Session B-07

Separating mass-mediated and non-mass-mediated effects on muscle strength using gene-level Mendelian randomization

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Muscle mass and strength are related but distinct phenotypes in sarcopenia. Contrary to the expectation that greater muscle mass should translate into greater strength, anabolic therapies such as bimagrumab can increase muscle mass without improvements in strength or physical function. This dissociation raises a question of whether the gene-level effect on strength is mediated by mass. Previous Mendelian randomization (MR) studies examined mass and strength separately, but parallel analyses do not resolve this distinction. Addressing this gap is important for sarcopenia target discovery, where strength effects should not be inferred from mass effects alone.

For each gene, we decomposed its effect on strength into mass-mediated and non-mass-mediated effects by two-step MR using GTEx v10 skeletal-muscle cis-eQTLs and UK Biobank appendicular lean mass and hand grip strength. The decomposition remains applicable to single-instrument genes, enabling genome-wide analysis. Across 10,853 genes, mass and strength effects correlated at r = 0.431 (attenuation-corrected r = 0.489), indicating substantial gene-level variation beyond a common mass–strength relationship. The mass-mediated effect largely restated the mass axis, whereas 4,370 genes (40.3%) showed a significant non-mass-mediated effect at FDR < 0.05. Varying the mass-to-strength slope preserved this set well above chance (median Jaccard = 0.811; random-set maximum = 0.274), supporting set-level robustness. Against 1,150 Mendelian muscle-disease genes, both effects remained independently associated with membership (OR 1.15 and 1.11 per SD; difference P = 0.49), supporting muscle-relevant information in the non-mass-mediated effect beyond the mass-mediated effect.

Gene effects on strength vary in the extent to which they are mediated by mass. The decomposition separates mass-mediated from non-mass-mediated effects at the gene level, although interpretation is limited by appendicular lean mass and grip strength as proxies for muscle mass and strength and by uncertainty in the mass-to-strength relationship, particularly for individual genes. Applied genome-wide, the framework identifies strength effects not captured by mass alone. With broader directional compound-phenotype annotations, these gene-level effects may support compound-level therapeutic prioritization.

Session B-08

Association Between Selective Reuptake Inhibitor Use and Reduced Cancer Risk in the UK Biobank

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Selective serotonin reuptake inhibitors (SSRIs) are widely prescribed for depression and anxiety. Emerging experimental evidence suggests that serotonin transporter inhibition may enhance antitumor immune activity, raising the possibility that SSRIs could reduce the risk of cancer. However, population-based evidence remains limited, particularly after accounting for psychiatric indications and related confounding factors. To evaluate the association between SSRI use and incident cancer risk in the UK Biobank and to examine whether this association differs according to individual SSRI agents and depression/anxiety status. We conducted a cohort study using UK Biobank data and applied propensity score matching to evaluate the association between SSRI use and incident cancer risk. SSRI users were matched with individuals without antidepressant exposure based on demographic, lifestyle, cardiometabolic, and psychiatric characteristics. Incident cancer was identified from cancer registry records. Cox proportional hazards models were used to evaluate the association between SSRI use and subsequent cancer risk, with additional analyses by individual SSRI and depression/anxiety status. SSRI use was associated with a significant reduction in the risk of incident cancer after propensity score matching. Inverse associations were also observed for selected SSRI agents. The association was more evident among participants without depression and among those without either depression or anxiety, suggesting that the findings were not solely attributable to psychiatric indications for SSRI treatment. This study provides population-based evidence supporting an association between SSRI use and reduced cancer risk. The findings strengthen the epidemiological basis for further investigation of serotonin transporter inhibition and serotonergic signaling as potential contributors to antitumor immunity and cancer prevention.

Session B-09

M-SCAN: Multi-Scale Constrained Analysis of Metabolic Networks

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Background
Cancer metabolism reflects both recurrent tumour programmes and tissue-specific constraints, but transcriptome analyses do not directly estimate metabolic reaction fluxes. We developed Multi-Scale Constrained Analysis of Metabolic Networks (M-SCAN), a reproducible workflow for sample-wise genome-scale flux inference across heterogeneous cancer cohorts.

Results
M-SCAN retains the Human1 genome-scale metabolic network as a shared reaction backbone and introduces sample specificity RNA expression, deleterious mutation effects, predicted kcat, direction-specific treatment of reversible reactions, and tissue-specific nutrient constraints. We benchmarked M-SCAN using measured exchange fluxes from 11 cancer cell lines and against an independent 13C-MFA dataset containing intracellular flux. Predicted and measured exchange fluxes were significantly correlated (Spearman $\rho = 0.704$, $P < 0.001$), and M-SCAN outperformed an enzyme constrained baseline in every cell line. We then applied M-SCAN to 7,562 tumour and normal samples from The Cancer Genome Atlas and the Genotype-Tissue Expression project across 14 cancer types. The flux profiles preserved tissue-specific metabolic structure, distinguished tumour from normal samples, and recovered flux programmes associated with clinical outcome and molecular subtype. Machine-learning analyses further identified recurrent tumour-associated pathways and stratified reactions into global-core, regional, and cancer-specific layers.

Conclusions
M-SCAN provides a scalable and reproducible framework for integrating molecular, enzymatic, and environmental constraints into comparable sample-specific flux states. The workflow enables cohort-scale analysis of shared and context-dependent metabolic programmes and provides reusable resources for hypothesis generation in cancer metabolism.

Session B-10

Information Extraction from Research Papers Using Sentence-based Chunking Combined with Keyword-based Sentence Extraction

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Retrieval Augmented Generation (RAG) system facilitates a unique opportunity to use both language models as well as various literature corpus together. This becomes an important aspect for information retrieval tasks as multiple documents and language models can be used at once to gather the required information. The use of multiple documents at once and generation of context to pass to the language model alongside the query remains a challenge. A high quality and information-heavy context can help produce accurate and relevant information from language models. We propose a method to first extract research-topic relevant sentences from a collection of research papers, then find the ones closest to the query and finally pass them as a context to the language model along with the query. We compare this method with the commonly used approach of passing a full pdf/document to generate context. Through our experiments using 10 research papers, we determined that our method produces accurate results (~70%) whilst taking considerably lower time (about 6 times faster than passing whole documents) and generating lower hallucinated results (~30%).

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